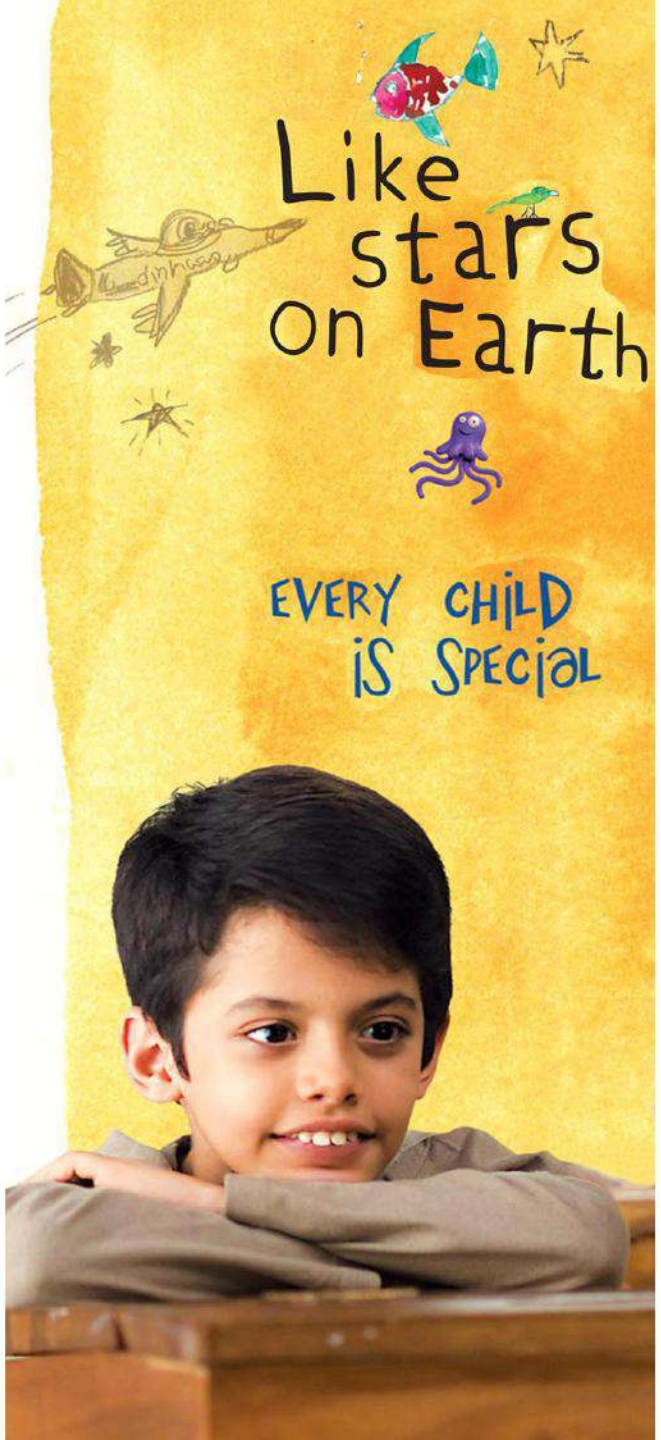


SCREENING AND REFINEMENT SYSTEM TO IDENTIFY THE RISK OF DYSCALCULIA AND DYSGRAPHIA LEARNING DISABILITIES IN PRIMARY SCHOOL STUDENTS

20_21-J 01

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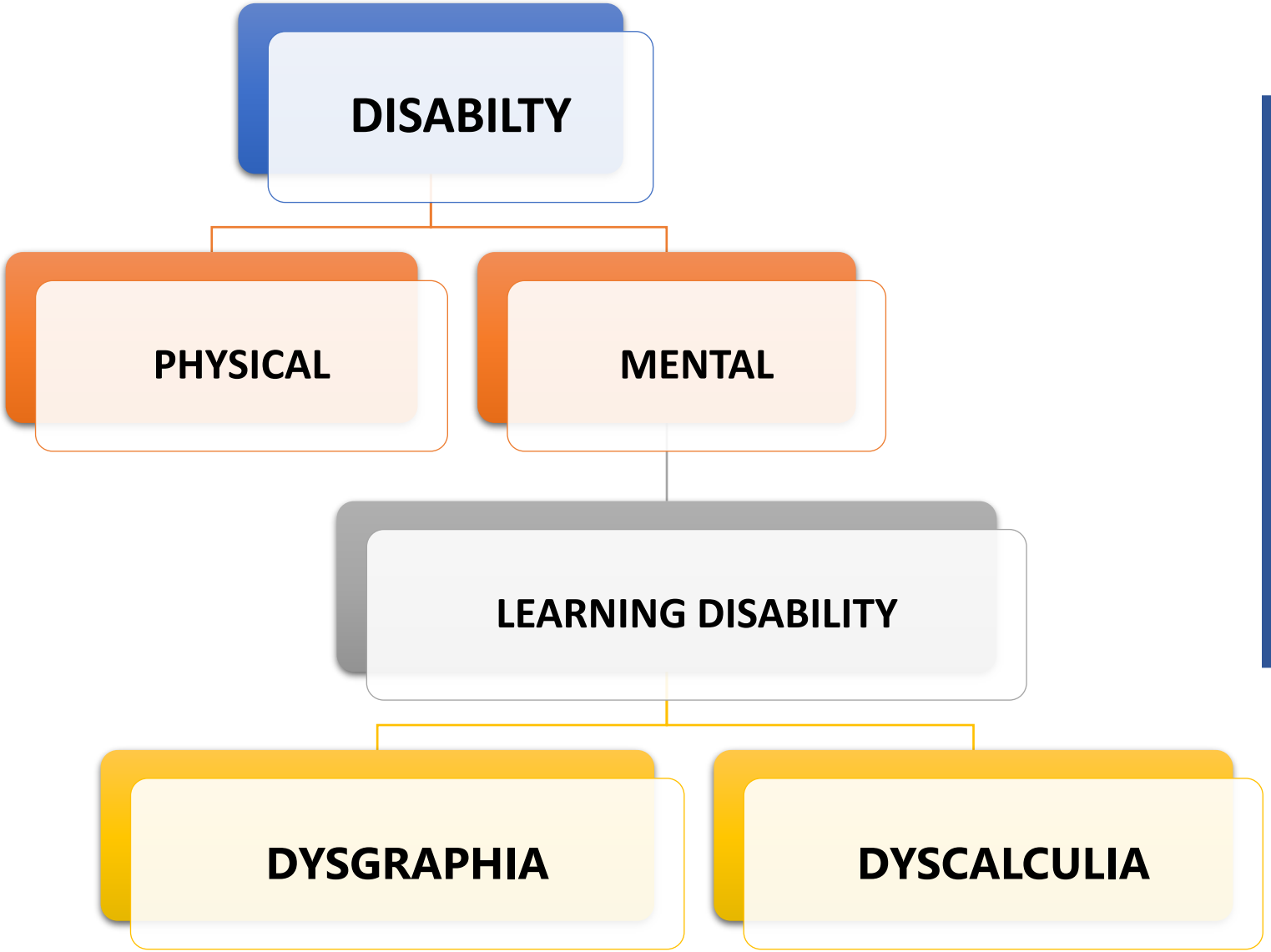


Liyanage T.J.P.S.

Content

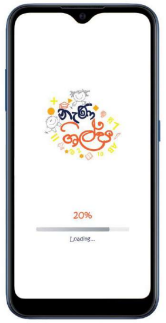
1. Introduction
2. Research Question
3. Objectives
4. System Overview Diagram
5. Technologies
6. Individual Components Section
7. Requirements
8. Gantt Chart
9. Commercialization Aspects
10. Budget

1. INTRODUCTION





INTRODUCTION CONT.



NANA SHILPA

DYSGRAPHIA

Difficulty of writing in provided space



Unclear and irregular handwriting



Slow writing pace



DYSCALCULIA



Mathematics disorder



Difficulty in identifying numbers



Difficulty in perform basic mathematical operations

2. RESEARCH QUESTION

How to provide mobile application-based solution to

- Screen and predict the risk of Dyscalculia and Dysgraphia learning disabilities(LD) in primary school students ?
- Deliver Comprehensive activities to overcome those LDs risks?





3. OBJECTIVES

MAIN OBJECTIVE

- Provide mobile application-based solution to screen and predict the risk of Dyscalculia and Dysgraphia learning disabilities in primary school students.

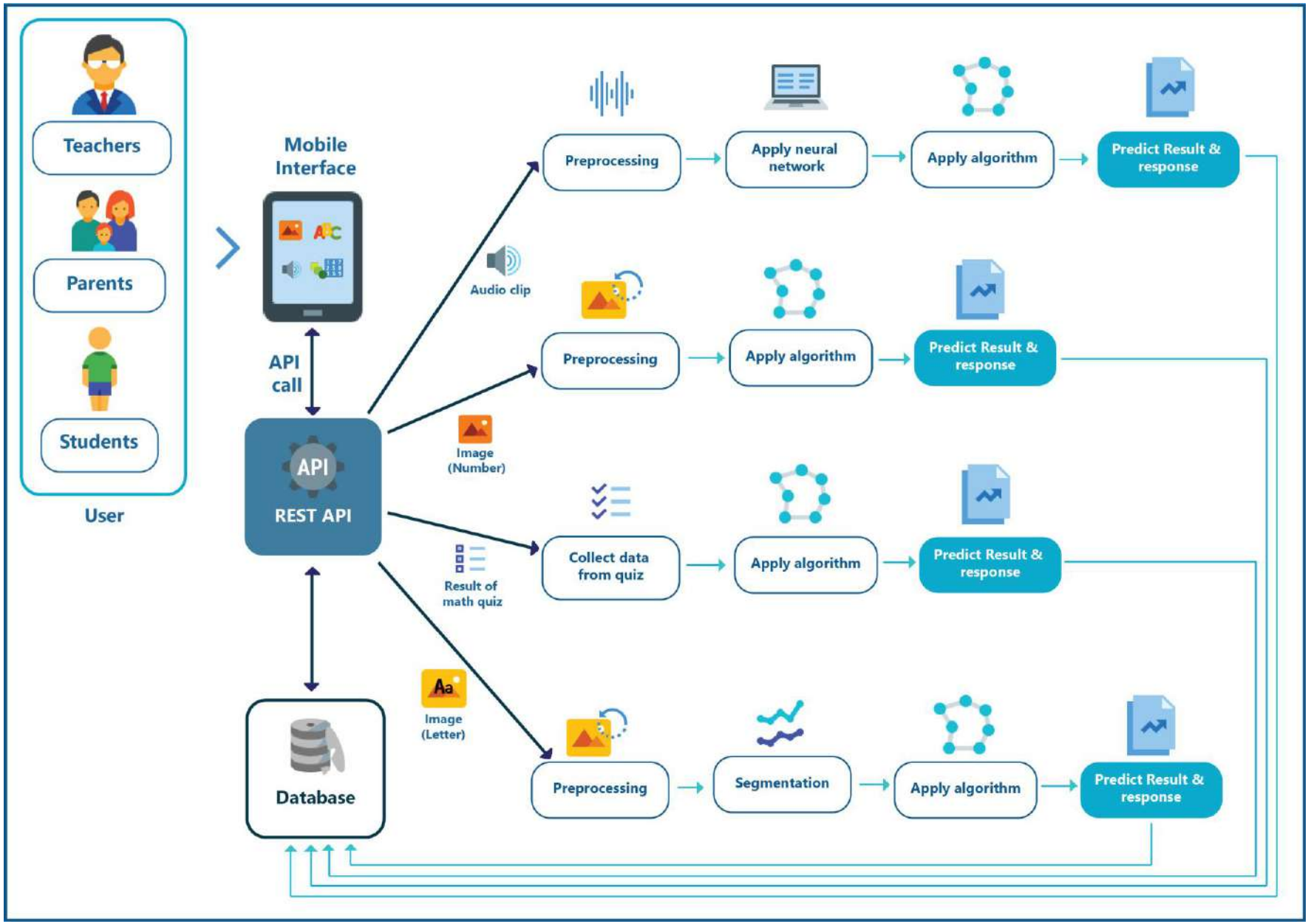


SPECIFIC OBJECTIVES



- Identify the risk of Letter level Dysgraphia through Sinhala Characters and provide activities to improve the condition.
- Identify the risk for Numeric Dysgraphia and provide activities to improve the condition.
- Identify the risk for Operational and Practognostic Dyscalculia and provide activities to improve the condition.
- Identify the risk for Lexical and Verbal Dyscalculia and provide activities to improve the condition.

4. SYSTEM OVERVIEW DIAGRAM



5. TECHNOLOGIES



Mobile Application

- Android

Database

- SQLite

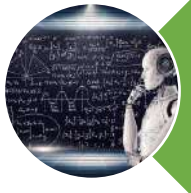
Middle Ware Technologies

- Python
- TensorFlow
- REST API - Flask



Technical Concepts

- Machine Learning
- Convolutional Neural Network (CNN)
- Image Processing



Algorithms

- Support Vector Machine
- FFT Algorithm



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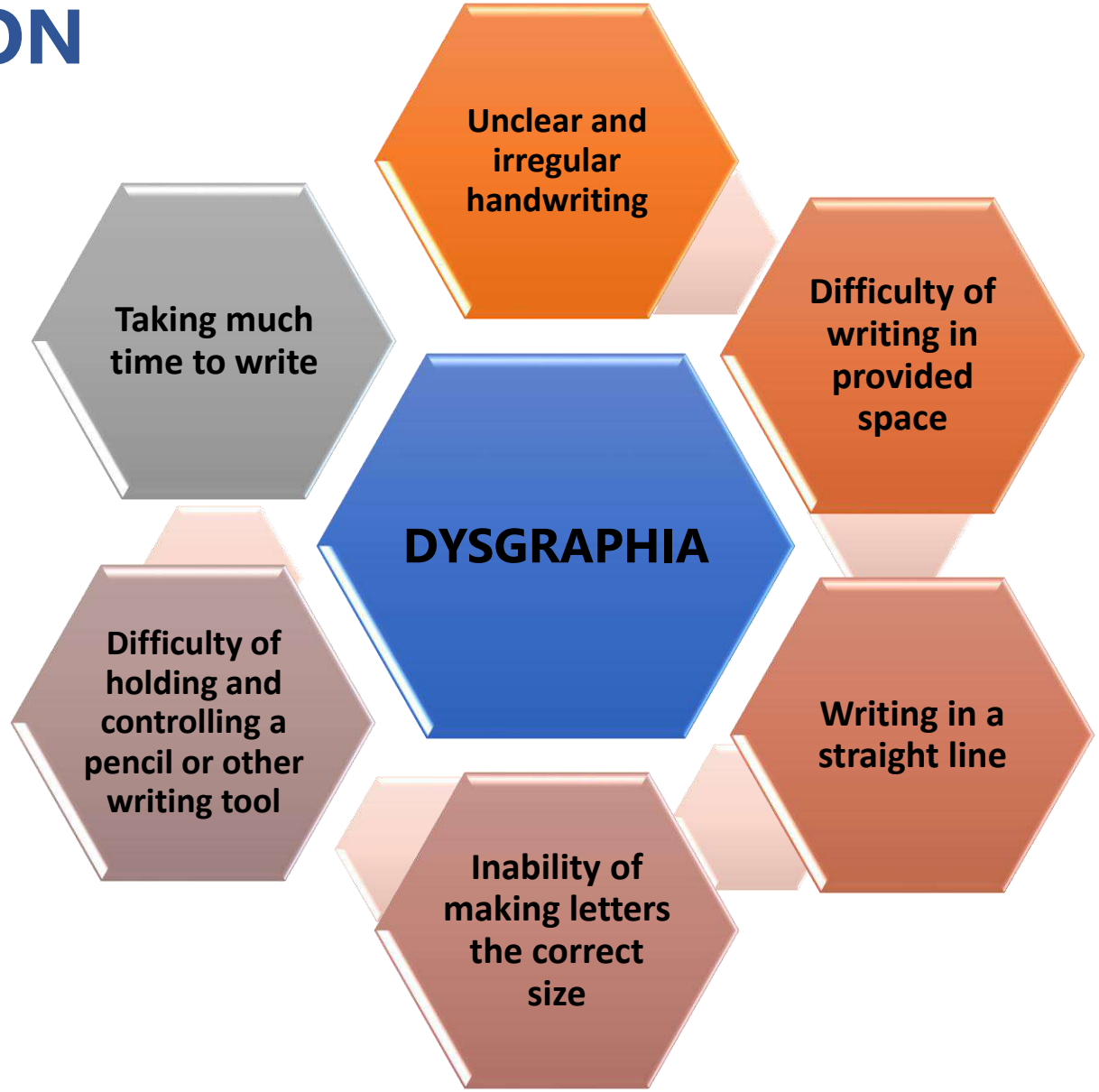
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COMPONENT 01

Identify the risk of Letter level Dysgraphia through Sinhala Characters and provide activities to improve the condition

❖ INTRODUCTION

❖ BACKGROUND



❖ RESEARCH GAP

Assistive Learning Environment for Students with Learning Disabilities[3]

- Proposed Assistive Learning Environment (ALE) for the students who are undergoing with learning disabilities.
- Provide environment for the students in order to follow the activities in their own pace.

A Mobile Web Game Approach for Improving Dysgraphia[4]

- Provide game-based solution for dysgraphia students.
- Implemented using client server web application architecture.

An Expert System for Diagnosing Dysgraphia [5]

- Identify the Dysgraphia condition through the decision tree analyze techniques.

Automatic segmentation as a tool for examining the handwriting process of children with Dysgraphic and proficient handwriting [6]

- System for the teachers and students where teacher can view all the class through on interface.
- Based on the initial conducted activities, improvement learning model is assigned by the teacher.

❖ SYSTEM COMPARISON

Features	01	02	03	04	Nena Shilpa
Screening and intervention for Dysgraphia	✗	✗	✗	✗	✓
Letter Identification	✗	✗	✗	✗	✓
Supports native Sinhala language	✗	✗	✗	✗	✓
Recover activities based on the risk level	✗	✗	✗	✓	✓
Keep track of the progress	✗	✗	✗	✗	✓
Report Generation	✗	✗	✗	✗	✓
User attractive interfaces	✗	✓	✗	✓	✓

01-Assistive Learning Environment for Students with Learning Disabilities[3]
02 -A Mobile Web Game Approach for Improving Dysgraphia[4]
03 -An Expert System for Diagnosing Dysgraphia [5]
04 -Automatic segmentation as a tool for examining the handwriting process of children with Dysgraphic and proficient handwriting [6]

❖ RESEARCH PROBLEM

How to provide mobile application-based solution to

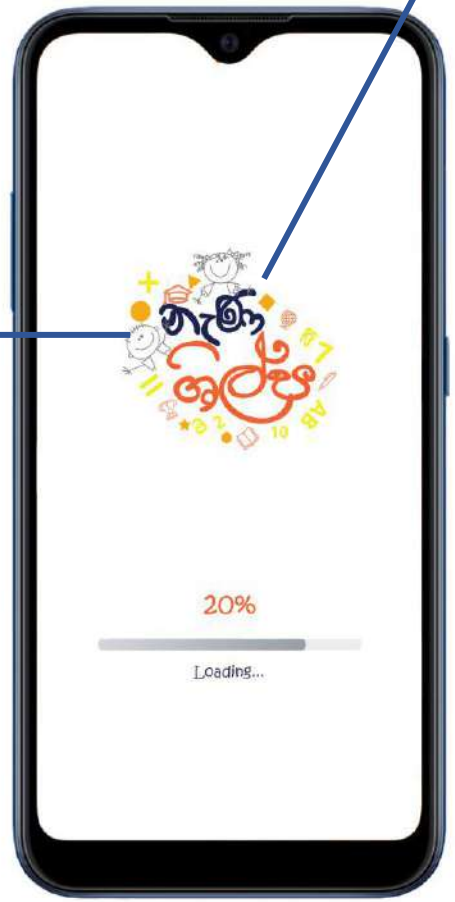
- **Identify the risk of letter level Dysgraphia condition ?**
- **Deliver risk diminish activities ?**
- **Monitor the betterment of the students ?**



❖ OBJECTIVES

SPECIFIC OBJECTIVE

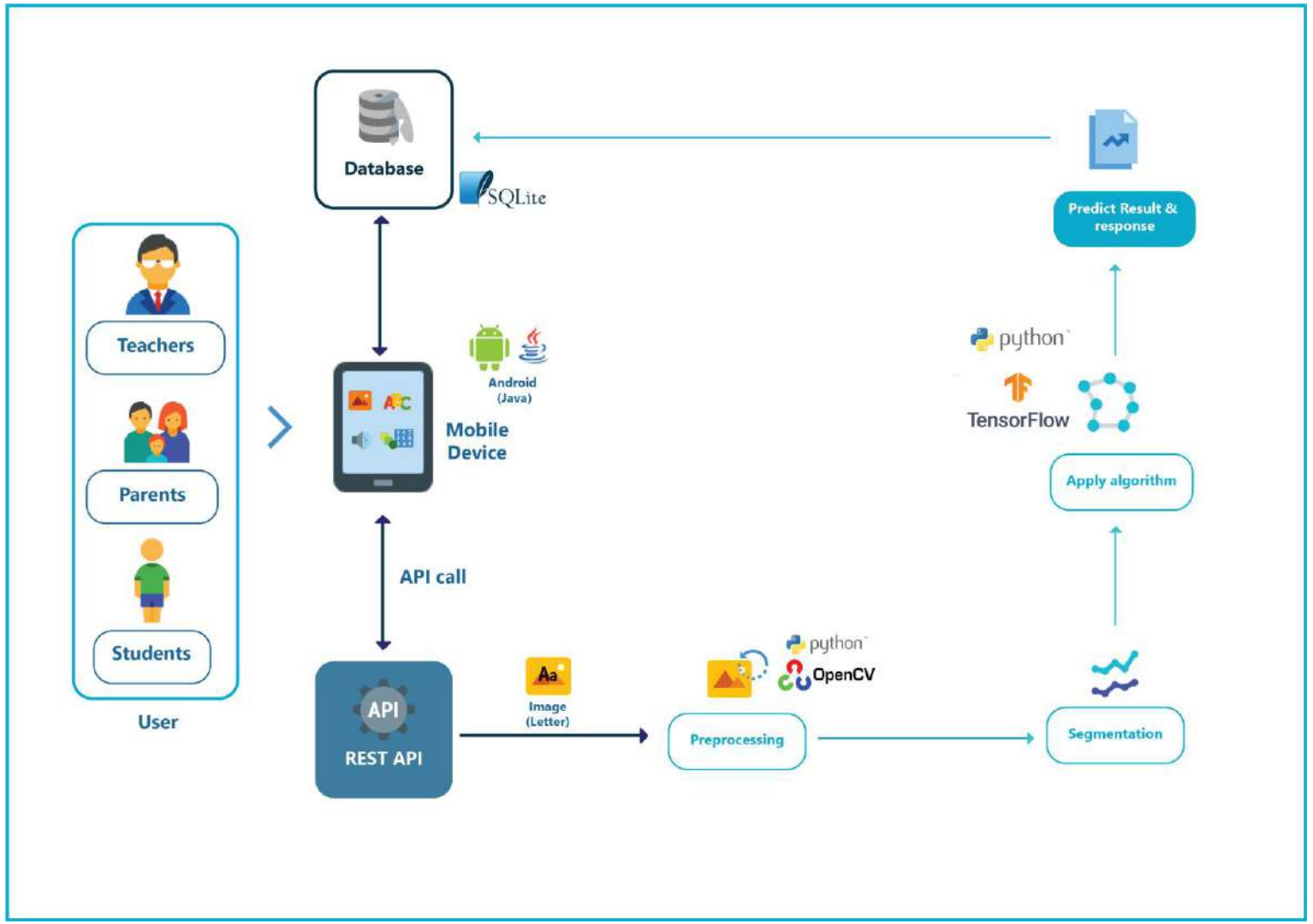
❖ Identify the risk of Letter level Dysgraphia through Sinhala Characters and provide activities to improve the condition.



Sub Objectives

- Letter level Dysgraphia risk identification.
- Practice activities to improve the condition.
- Keep the track of student activities.
- Report generation for the performed test activities.

METHODOLOGY



Component Overview Diagram

❖ KEY PILLARS


Reinforcement Learning
Trained Data Model
Machine Learning
Conventional Neural Network
Image Processing

❖ COMPONET SPECIFIC REQUIREMENTS

FUNCTIONAL REQUIREMENTS



User Login to application



Take the risk prediction test for letter level dysgraphia



View the prediction test results



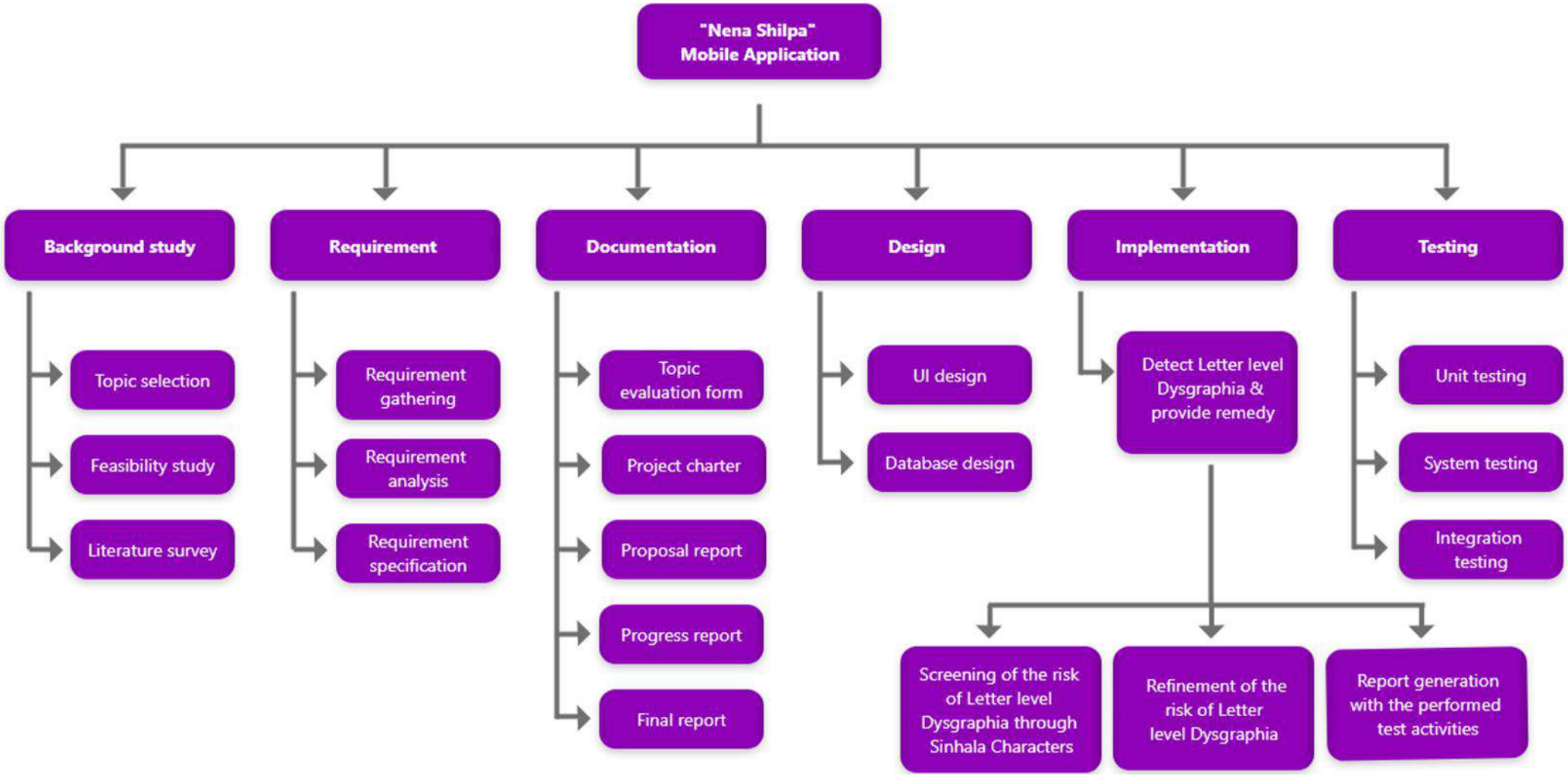
Suggest the activities for condition improvement



View Progress Report



❖ WORK BREAKDOWN CHART



❖ REFEREANCES

[01]Dysgraphia Information Page. (n.d.). Retrieved July 21, 2020, from https://www.ninds.nih.gov/Disorders/All-Disorders/Dysgraphia-Information-Page

[02]Mishna, F. (2003). Learning Disabilities and Bullying. *Journal of Learning Disabilities, 36*(4), 336-347. doi:10.1177/00222194030360040501

[03]Z. Pirani and M. Sasikumar, "Assistive learning environment for students with Learning Disabilities," *2012 4th International Conference on Intelligent Human Computer Interaction (IHCI)*, 2012

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[04] D. Giordano and F. Maiorana, "A Mobile Web Game Approach for Improving Dysgraphia," Proceedings of the 7th International Conference on Computer Supported Education, 2015.

[05] D. A. Kurniawan, S. W. Sihwi, and Gunarhadi, "An expert system for diagnosing dysgraphia," 2017 2nd International conferences on Information Technology, Information Systems and Electrical Engineering (ICITISEE), 2017.

[06] Rosenblum, Sara & Dvorkin, Assaf & Weiss, Patrice. (2006). Automatic segmentation as a tool for examining the handwriting process of children with dysgraphic and proficient handwriting. Human movement science. 25. 608-21. 10.1016/j.humov.2006.07.005.



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COMPONENT 02

Identify the risk for Numeric Dysgraphia and provide activities to improve the condition.

INTRODUCTION

❖ BACKGROUND

- A special Learning Disability called **Numeric Dysgraphia (ND)** not a very common term.
- This disability causes difficulties when writing the numbers.
- No special tests or treatments are not identified and cannot be cured [1].
- This kind of students can be identified by their numeric writing patterns by the teacher or a parent who pays much attention.



Through activities / practices, the situation of the student can be improved to some extent. So, it is important to identify this kind of students at early stage.

RESEARCH GAP

IO-CRIVO mobile app[1]

- For the students who has difficulties in writing.
- Screening for Dysgraphia
- PHP, HTML, CSS, JavaScript, MySQL

An Expert System[2]

- Provide remedy for Dysgraphic students on their studies
- A writing task and questionnaire was provided.

AR-DAWE model[3]

- AR based Dysgraphia Assistance Writing Environment (AR-DAWE).
- Identify spelling mistakes of the students.

A writing questionnaire[4]

- A statistical model
- A questionnaire

SYSTEM COMPARISON

Features	[1]	[2]	[3]	[4]	Nena Shilpa
Screening or intervention for Dysgraphia	✓	✓	✓	✓	✓
Screening and intervention for Dysgraphia	✗	✗	✗	✗	✓
Screening for Numeric Dysgraphia	✗	✗	✗	✗	✓
Remedies for ND in activity-based environment	✗	✗	✗	✗	✓
Keep track of the progress	✓	✗	✗	✗	✓
Support for Sinhala Language	✗	✗	✗	✗	✓
User attractive interfaces with sound bases	✗	✗	✓	✗	✓

RESEARCH PROBLEM

How to develop a solution which is a mobile application,



To identify the risk for ND ?



To provide remedies ?



To keep the track of progress ?



OBJECTIVES

SPECIFIC OBJECTIVE

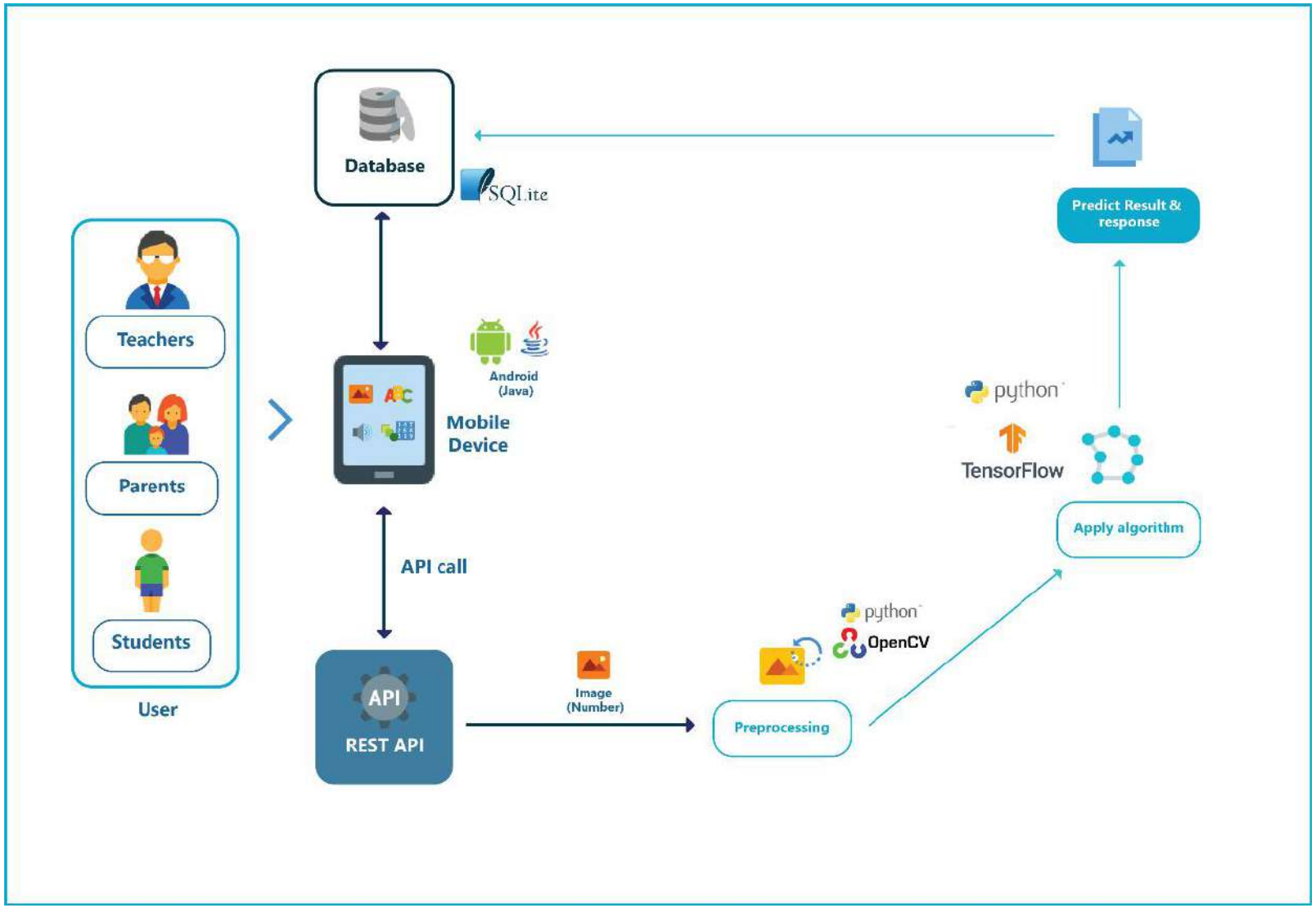
- Identify the risk for Numeric Dysgraphia and provide remedies while keeping the track of progress.



Sub Objectives

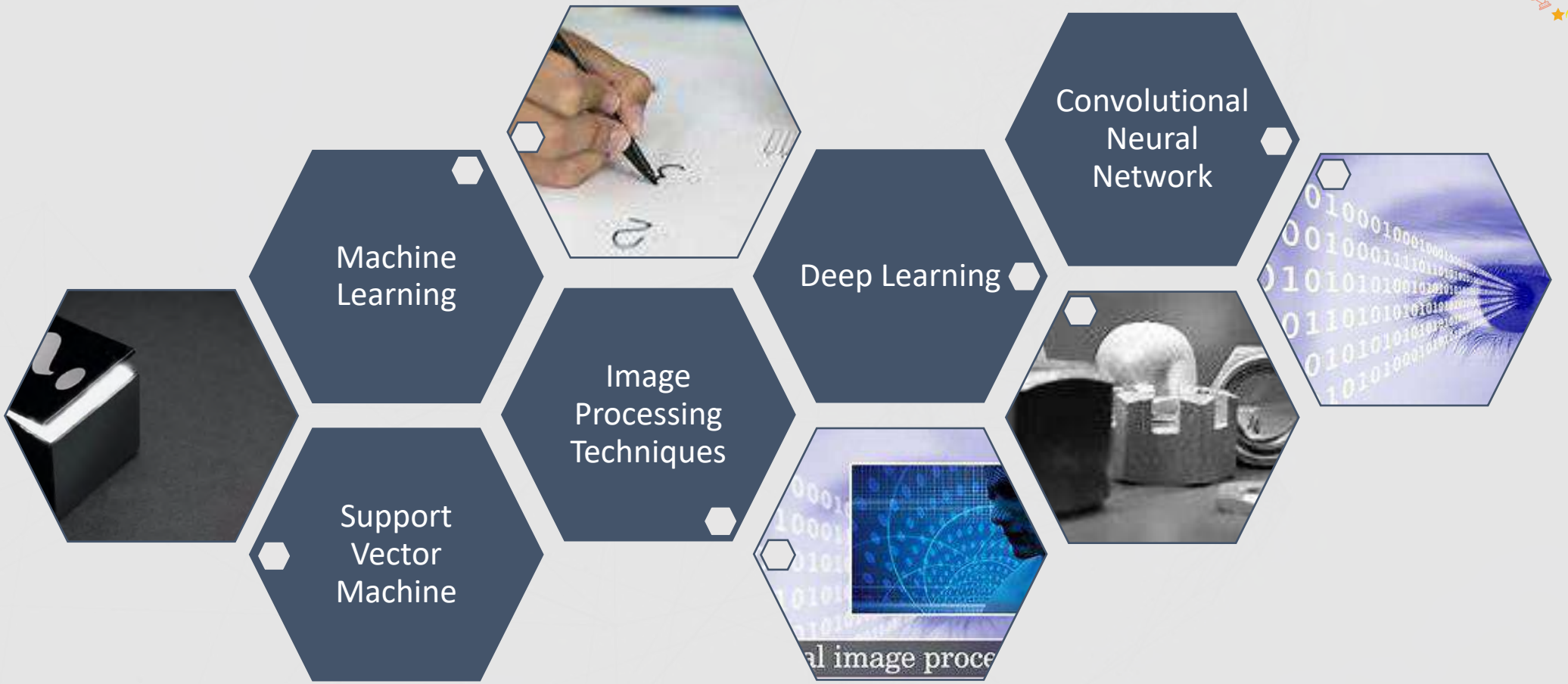
- Screening for the risk for Numeric Dysgraphia
- Providing activity-based treatments
- Keep the track of progress

METHODOLOGY



System Component Overview Diagram

KEY PILLARS


Machine Learning
Deep Learning
Convolutional Neural Network
Image Processing Techniques
Support Vector Machine

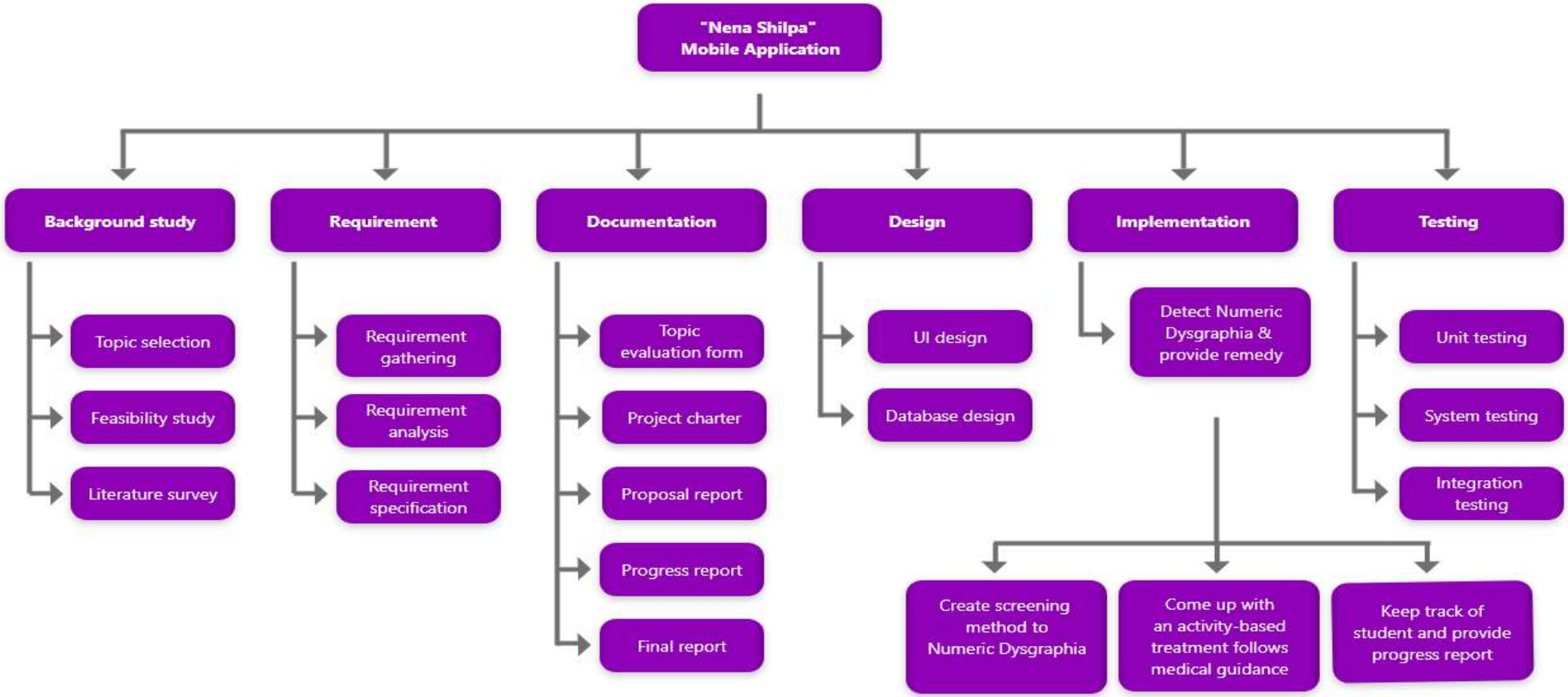
COMPONENT SPECIFIC REQUIREMENTS

Functional Requirements

- User Login
- Screening the risk for ND
- Providing Remedies
- Keeping the track of progress



❖ WORK BREAKDOWN CHART – Numeric Dysgraphia



❖ REFERENCES

[1]. Giordano, D. & Maiorana, Francesco. (2014). Addressing dysgraphia with a mobile, web-based software with interactive feedback. 2014 IEEE-EMBS International Conference on Biomedical and Health Informatics, BHI 2014. 264-268. 10.1109/BHI.2014.6864354.

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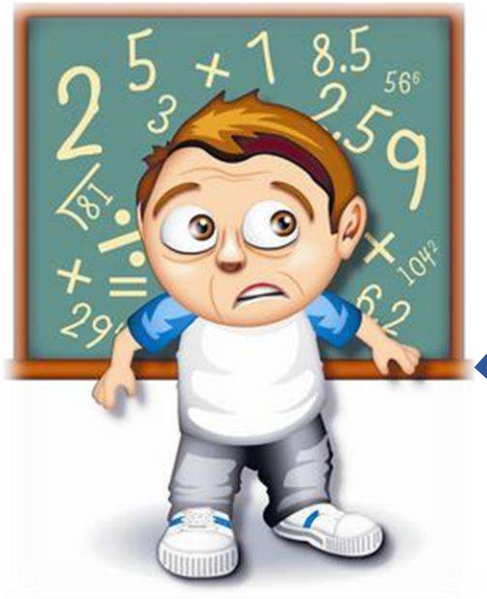
COMPONENT 03

Identify the risk for Operational and Practognostic Dyscalculia and provide activities to improve the condition.

INTRODUCTION

❖ Background

- ❖ Dyscalculia or mathematics learning disability it is a specific learning disability directed to arithmetic.[1][2].
- ❖ Dyscalculia is sometimes called as number blindness.
- ❖ The prevalence of developmental dyscalculia is between 3% and 6% in the school population.
- ❖ Among the dyscalculia types two of them are focusing on here,
  - **Operational Dyscalculia**
  - **Practognostic Dyscalculia**



RESEARCH GAP

Screening for Disorders of Mathematics via a web application [3]

- Learning process with calculation, numeric concepts, and problem solving.
- HTML5, CSS, JavaScript while using the MySQL as the database.

Development of Assistive Technology for Students with Dyscalculia [4]

- Provided Solution as a software tool which named as "Calculating Aid Tools: KidKanit" which helps to dyscalculia students in mathematical area.

Web-based learning tool for primary school student with Dyscalculia [5]

- Cover the most common characteristics which were the Dyscalculia students struggle with in specific areas such as counting, number recognition.

Screening for Dyscalculia within HE (Higher Education) [6]

- Provide screening process to identify dyscalculia in higher education students.
- In their solution, it was made a mobile app called DyscalculiUM.

SYSTEM COMPARISON

Features	[3]	[4]	[5]	[6]	Nena Shilpa
Dyscalculic Identification.	✗	✗	✗	✓	✓
Provide activities to improve the math concepts.	✓	✓	✓	✗	✓
Support for Sinhala Language	✗	✗	✗	✗	✓
User attractive interfaces with sound bases (multi-scenery)	✗	✓	✗	✗	✓
Keep track of progress	✗	✗	✗	✗	✓
Report Generation	✗	✗	✗	✗	✓
Supervised Learning(uses Machine Learning)	✗	✗	✗	✗	✓

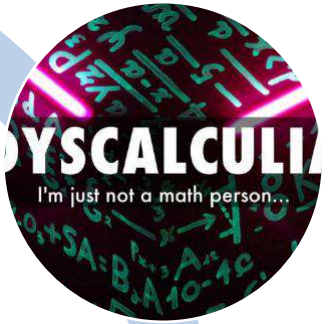




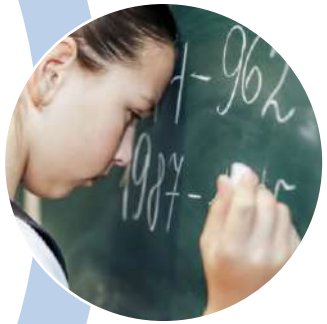
- Screening for Disorders of Mathematics via a web application [3]
- Development of Assistive Technology for Students with Dyscalculia [4]
- Web-based learning tool for primary school student with Dyscalculia [5]
- Screening for Dyscalculia within HE (Higher Education) [6]

RESEARCH PROBLEM

How to develop mobile **application-based solution to**



- Identify the risk for Operational and Practognostic Dyscalculia ?



- Provide activities to improve the condition ?



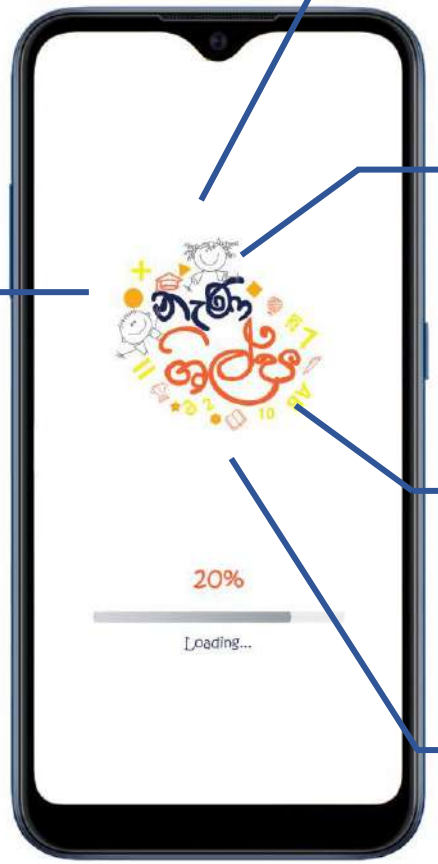
- Keep the track of progress ?



OBJECTIVES

SPECIFIC OBJECTIVE

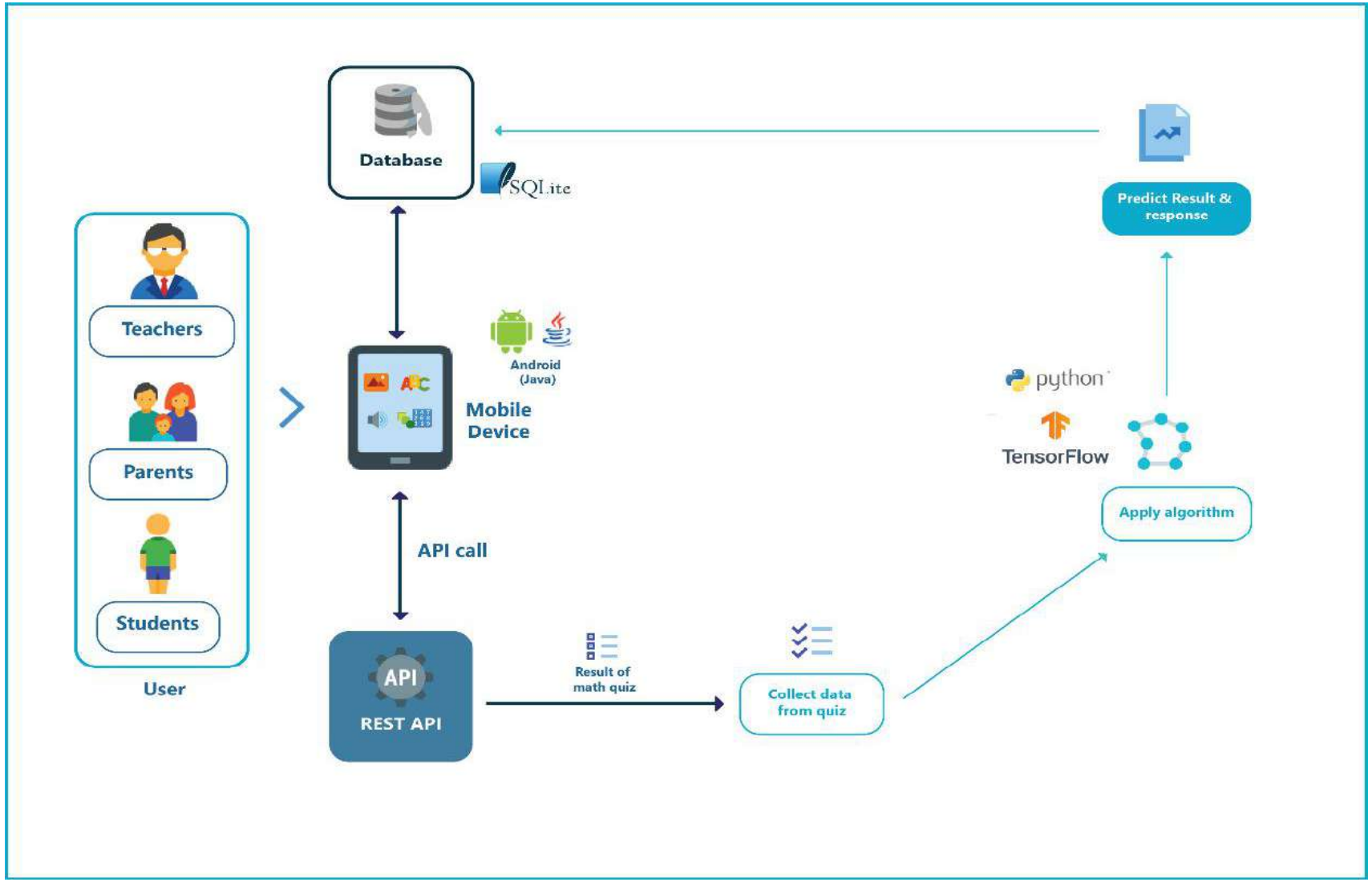
- Identify the risk for Operational and Practognostic Dyscalculia and provide activities to improve the condition.



Sub Objectives

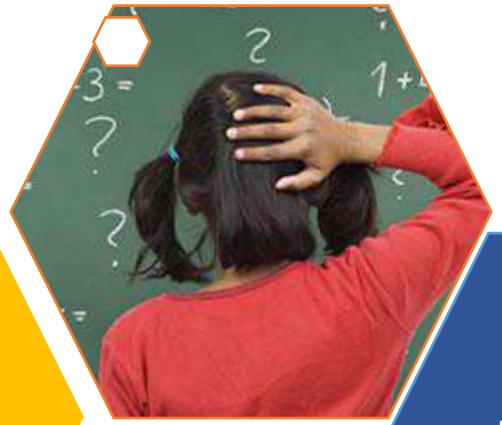
- Identify risk of Operational and Practognostic Dyscalculia.
- Provide activities for practice to improve the condition.
- Keep the track of student progress.
- Create Screening model

RESEARCH METHODOLOGY



Component Overview Diagram

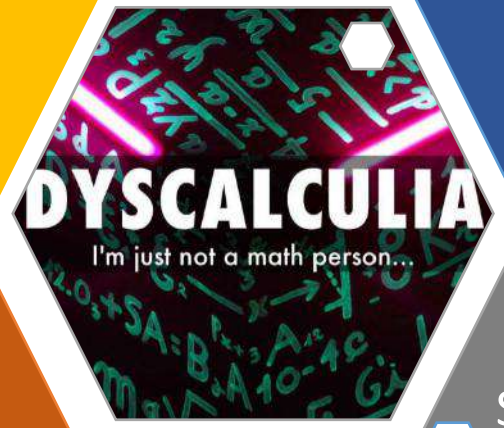
KEY PILLARS



Machine Learning

Data mining







Supervised Learning

Support Vector Machine

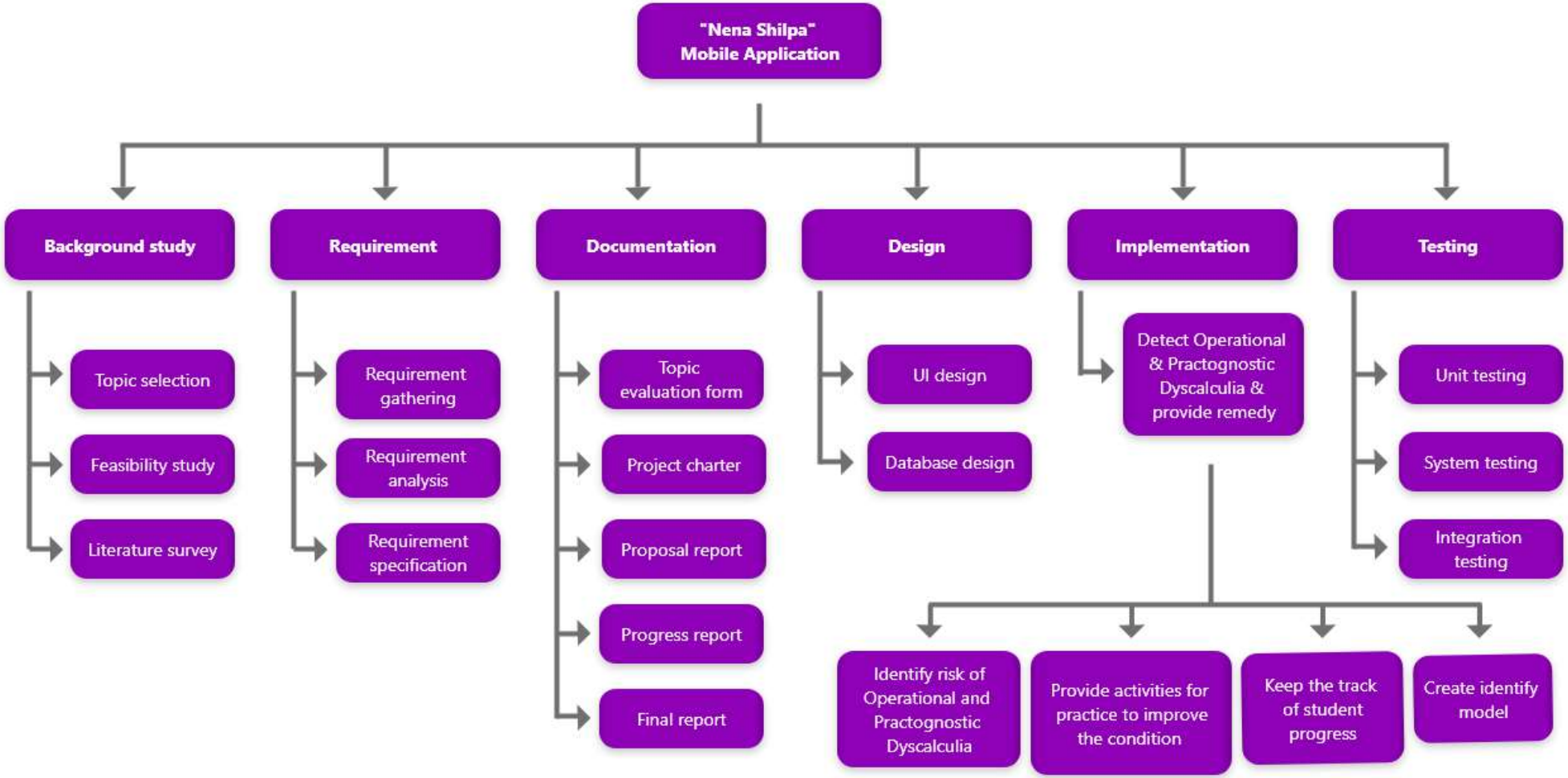
❖ COMPONENT SPECIFIC REQUIREMENTS

❖ Functional requirement

- ❖Login to the system
- ❖ Attempt the dyscalculia risk identification test.
- ❖View the quiz result
- ❖Provide training activities to overcome the symptoms.
- ❖Keep track of student progress



❖ WORK BREAKDOWN STRUCTURE



❖ REFERENCES

[1] Dysgraphia Information Page. (n.d.). Retrieved July 21, 2020, from [Online]. Available:https://www.ninds.nih.gov/Disorders/All-Disorders/Dysgraphia-Information-Page [Accessed: 26-July- 2020]

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COMPONENT 04

Identify the risk for Lexical and Verbal Dyscalculia and provide activities to improve the condition

INTRODUCTION

BACKGROUND

- Dyscalculia is a biologically based learning disability related to mathematic, numbers, and arithmetic skills.
- Centered around Neurological system.
- It is generally seen as mathematical equivalent of Dyslexia.
- According to British Dyslexia Association, Dyscalculia occur in 3 - 6% of population [1].
- There are 5 main types of Dyscalculia [2].
- Two types of them are focusing on here,

1. **Verbal Dyscalculia**
2. **Lexical Dyscalculia**



RESEARCH GAP

Features	1	2	3	4	Nena Shilpa
Number identification	✓	✓	✓	✓	✓
Voice recognition techniques	✗	✗	✗	✗	✓
Support Sinhala Language	✗	✗	✗	✗	✓
Improved UI/UX	✓	✓	✗	✓	✓
Keep track of progress	✗	✗	✗	✗	✓
Report Generation	✗	✓	✗	✗	✓
ML techniques	✗	✗	✗	✗	✓





1 - Mobile App for Dyscalculia Children in Malaysia [3]

2 - Mobile App Development on Detection and Diagnose of Learning Disability for Children [4]

3 - Intelligent System for Diagnosing Learning Disorders in Children [5]

4 - KidKanit

RESEARCH PROBLEM

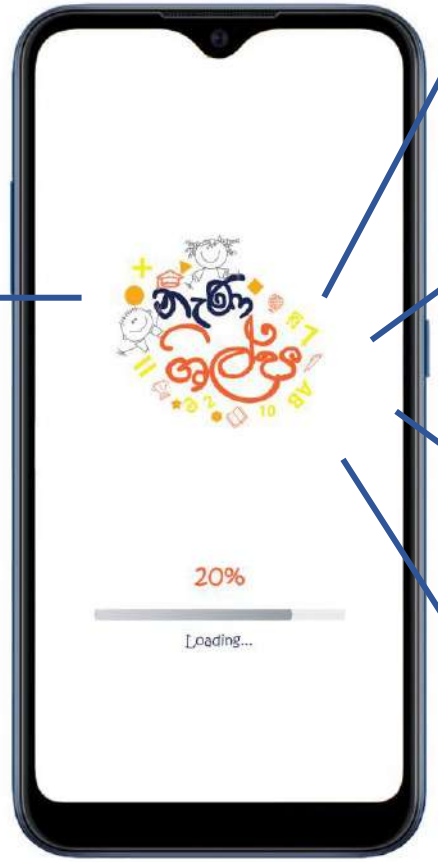


- How to Identify the risk for Lexical and Verbal Dyscalculia ?
- How to Provide activities to improve the condition ?
- How to Keep the track of progress ?

OBJECTIVES

Specific Objective

Identify the risk of Lexical and Verbal Dyscalculia and provide activities to improve the condition

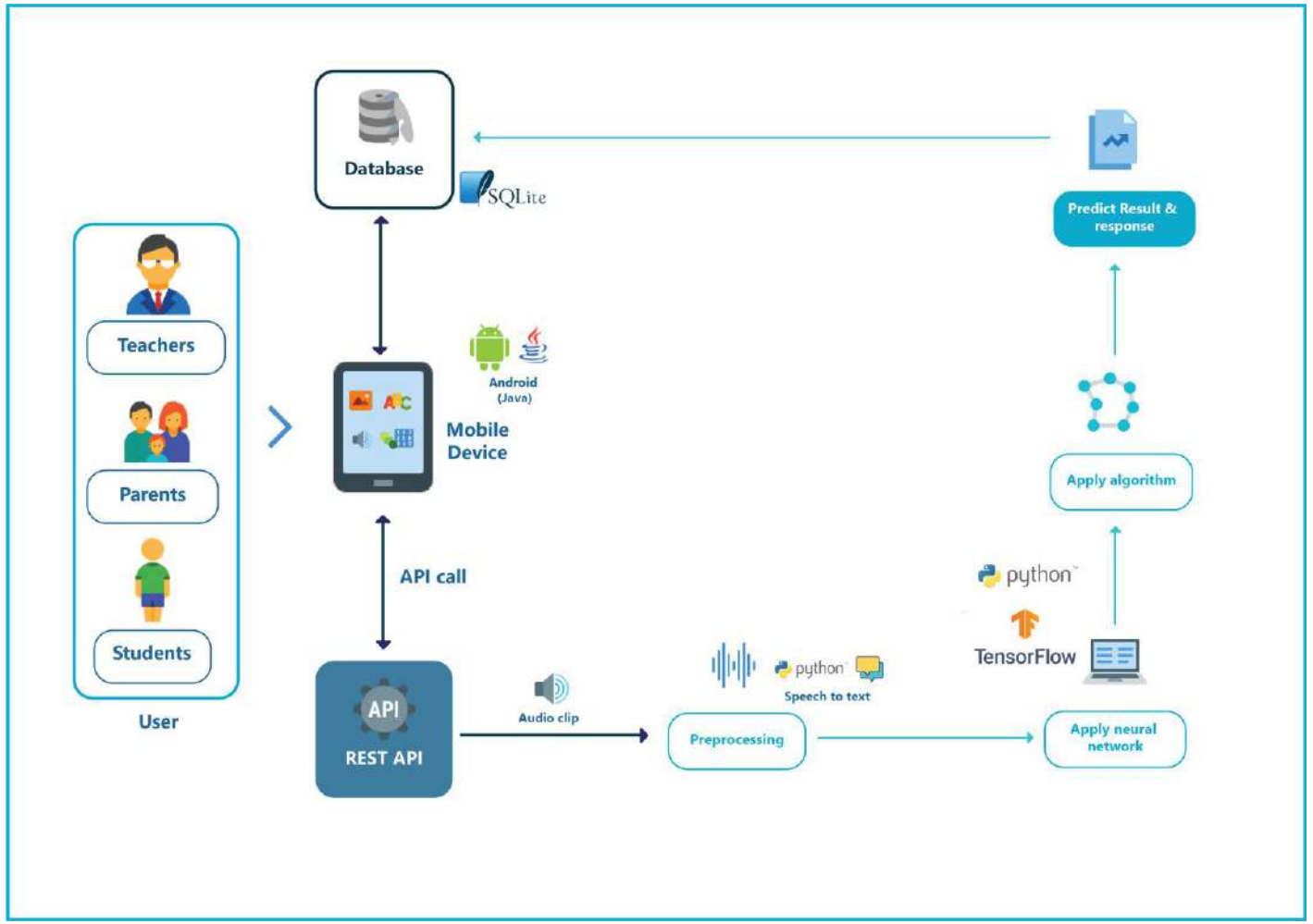


Sub Objectives

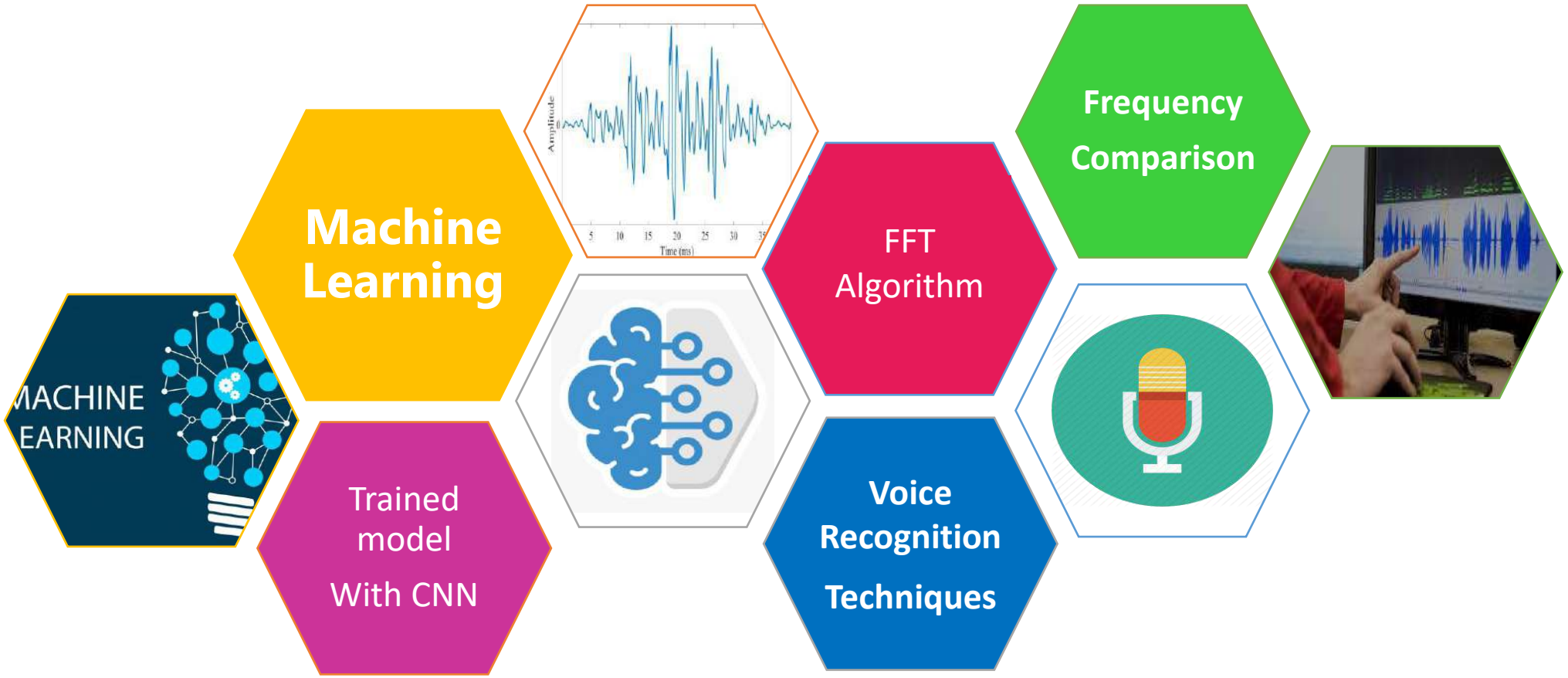
- Identify risk of Lexical Dyscalculia.
- Identify risk of Verbal Dyscalculia.
- Generate report with performed results
- Suggest remedies for identified user with risk.

METHODOLOGY

COMPONET OVERVIEW DIAGRAM



KEY PILLARS


Machine Learning
MACHINE LEARNING
Amplitude
Time (ms)
FFT Algorithm
Frequency Comparison
Trained model With CNN
Voice Recognition Techniques

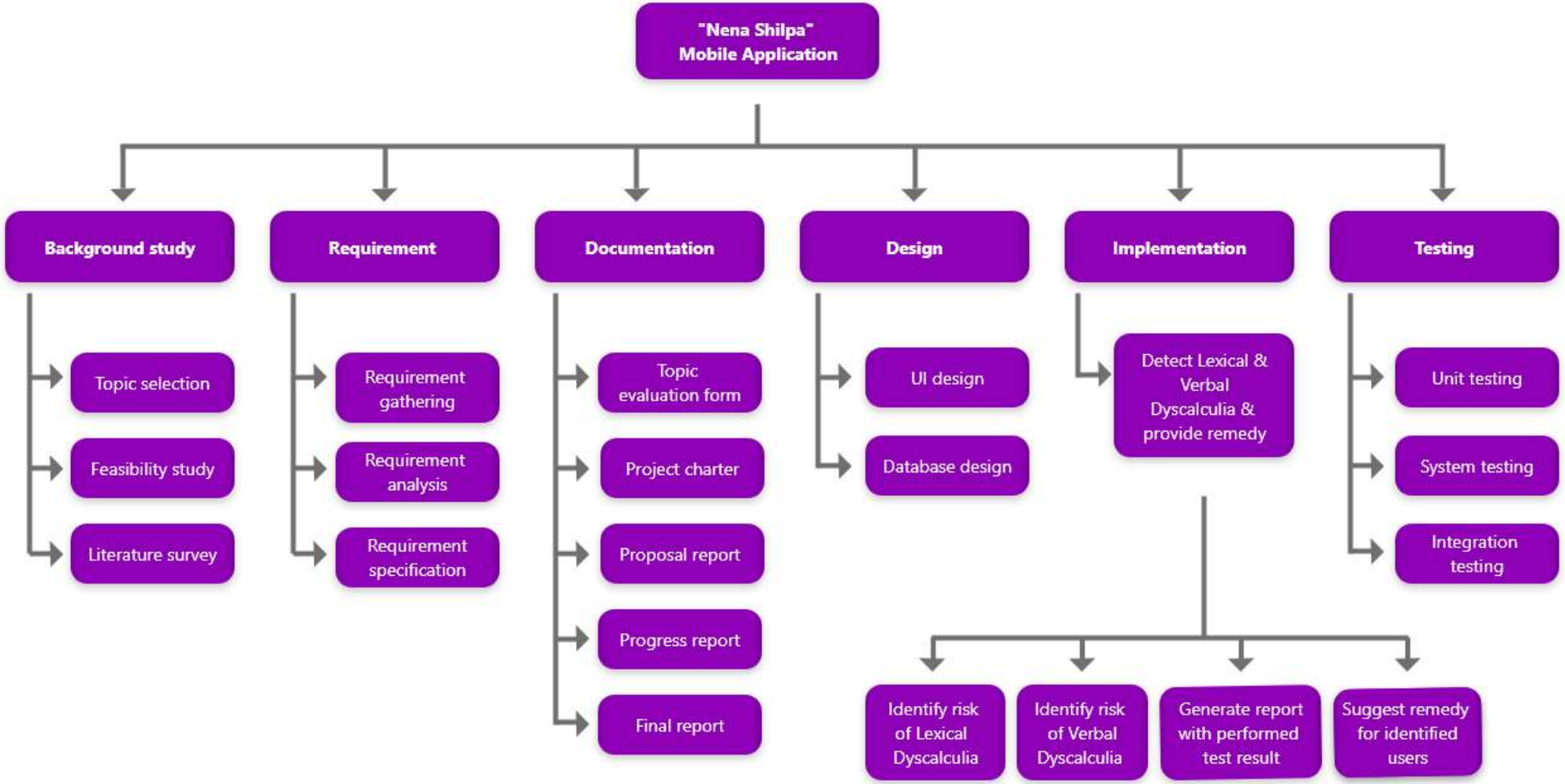
COMPONET SPECIFIC REQUIREMENTS



Functional Requirements.

- Provide activities to identify Lexical and Verbal Dyscalculia.
- Recognize voice inputs.
- Play recorded voice.
- Generate reliable report.
- Provide remedies.

WORK BREAKDOWN CHART



REFERENCES

[1] "Number Dyslexia: Everything to know about Dyscalculia - Number Dyslexia", *Number Dyslexia*, 2020. [Online]. Available: https://numberdyslexia.com/dyscalculia-aka-number-dyslexia/

[2] "CogniFit", *Beating dyscalculia: Treatment, Exercises, Causes, Symptoms, Types, Assessment and Definition*, 2020. [Online]. Available: https://www.cognifit.com/pathology/dyscalculia.

[3] M. Mohd Ariffin, F. A. Abd Halim, and N. Abd Aziz, "Mobile application for dyscalculia children in Malaysia," *Proc. 6Th Int. Conf. Comput. Informatics Embrac. Eco-Friendly Comput.*, no. April, pp. 467–472, 2017, [Online]. Available: http://www.uum.edu.my.

[4] G. Rajivsureshkumar, K. Malarvizhi, and G. Deebanchakkarawarthi, "Mobile Application Development on Detection and Diagnose of Learning Disability for Children," no. 6, pp. 216–224, 2019.

[5] M. A. J. Member and B. Shilpa, "Intelligent System for Diagnosing Learning Disorders in Children," *Lect. Notes Eng. Comput. Sci.*, vol. 2186, no. 1, pp. 70–75, 2010.



7. REQUIREMENTS

Personal Requirements

1. Students
2. Teachers
3. Parents
4. Medical officer



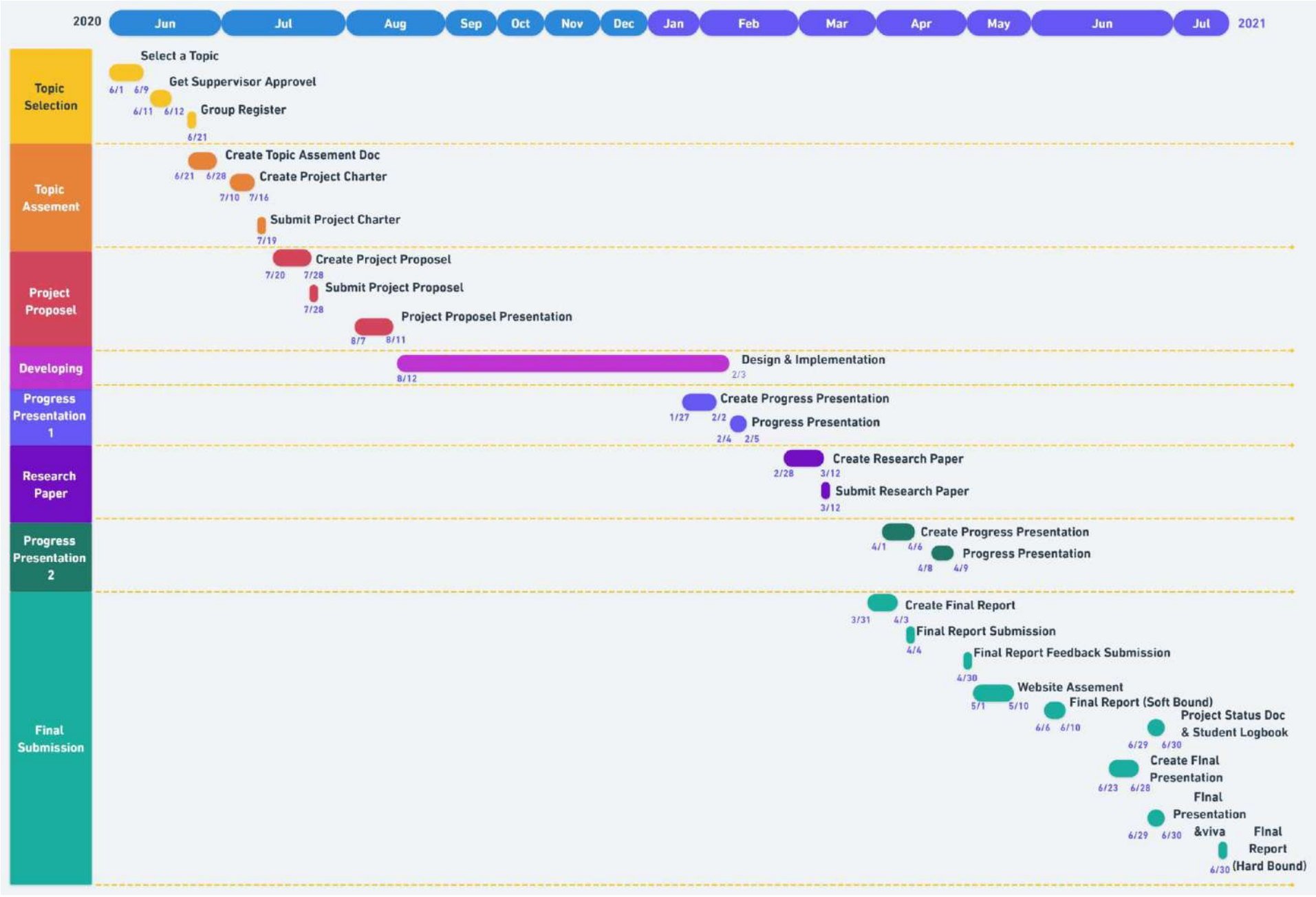
System Requirements

- ❑Software
  - **User-end**
    - ➢Android version 5 or above
  - **Developer-end**
    - ➢Android studio
    - ➢TensorFlow v2.0
    - ➢Python v3.7
    - ➢SQLite
- ❑Hardware
  - ➢Speaker
  - ➢ Microphones
  - ➢User Mobile Touch Screen

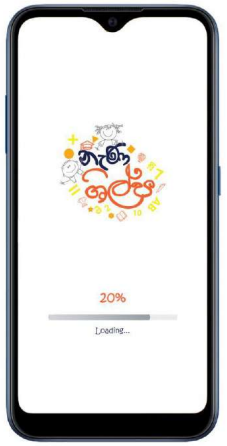
Non-functional Requirements

- ➢Reliability
- ➢Security
- ➢Performance
- ➢Usability
- ➢Availability

8. GANNT CHART



9. COMMERCIALIZATION ASPECTS



Using the proposed application, we hope to provide low-cost subscription plans considering each side.



For Schools
Rs. 1500

Permanent Subscription
Rs. 1000

Monthly Subscription
Rs. 300

Annual Subscription
Rs. 600

Low income Families.
Free Subscription.

Child Orphanages.
Free Subscription

Social Media Commercialization.

10. BUDGET

Component	Amount (USD)	Amount (LKR)
Traveling fee for the possible consultation sessions and data gathering (Malabe, School selected for data extracting)	1.08	200.00
Internet charges (the development and technical information learning)	10.77	2000.00
Technical consultation charges (External technical information session)	5.4	1000.00
Total	17.25	3200.00

APPENDIX



To whom it may concern, 19 July 2020

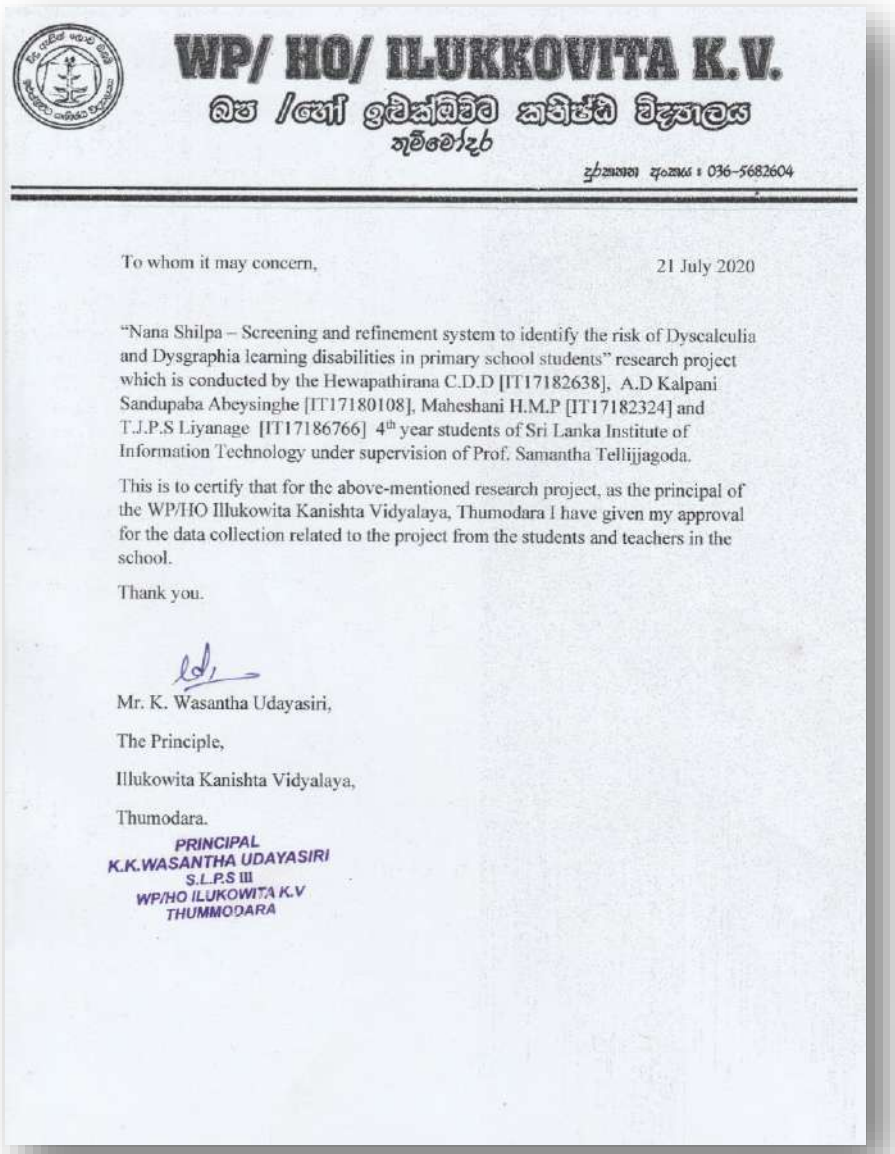
"Nana Shilpa – Screening and refinement system to identify the risk of Dyscalculia and Dysgraphia learning disabilities in students" research project which is conducted by the Hewapathirana C.D.D [IT17182638], A.D Kalpani Sandupaba Abeysinghe [IT17180108], Maheshani H.M.P [IT17182324] and T.J.P.S Liyanage [IT17186766] 4th year students of Sri Lanka Institute of Information Technology under supervision of Prof. Samantha Tellijjagoda.

This is to certify that for the above-mentioned research project, I will be providing the medical consultation for all the medical related aspects in this project. I hereby confirm that as an external supervisor of the project, I will be offering my consultation as a medical officer throughout this project.

Thank you.

Dr. Kamalani Wanigasinghe

Dr. W. M. K. WANIGASINGHE
(M.B.B.S.)
S.L.M.C. Reg. No: 11614
MALABE



WP/ HO/ ILUKKOVITA K.V.
ඔබ /ගෝ ඉඹක්ඛිට්ට කනිජ්ඨ විදුහාලය
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දුරකතන අංකය : 036-5682604

To whom it may concern, 21 July 2020

"Nana Shilpa – Screening and refinement system to identify the risk of Dyscalculia and Dysgraphia learning disabilities in primary school students" research project which is conducted by the Hewapathirana C.D.D [IT17182638], A.D Kalpani Sandupaba Abeysinghe [IT17180108], Maheshani H.M.P [IT17182324] and T.J.P.S Liyanage [IT17186766] 4th year students of Sri Lanka Institute of Information Technology under supervision of Prof. Samantha Tellijjagoda.

This is to certify that for the above-mentioned research project, as the principal of the WP/HO Illukowita Kanishta Vidyalaya, Thumodara I have given my approval for the data collection related to the project from the students and teachers in the school.

Thank you.

Mr. K. Wasantha Udayasiri,

The Principle,

Illukowita Kanishta Vidyalaya,

Thumodara.

PRINCIPAL
K.K.WASANTHA UDAYASIRI
S.L.P.S III
WP/HO ILUKOWITA K.V
THUMMODARA





Q & A.



THANK YOU